

Embracing Data-Centricity

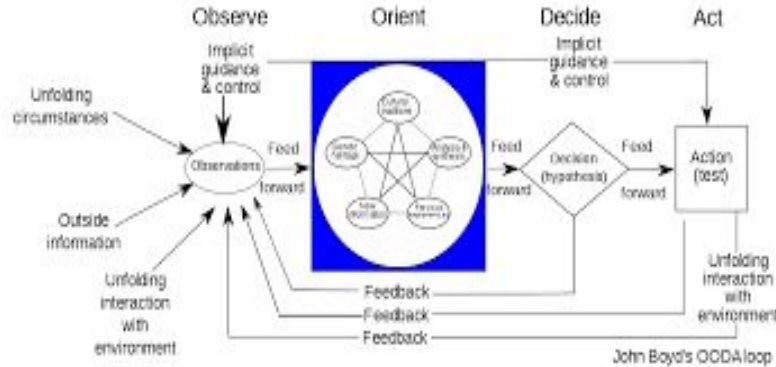
New Thinking, Planning, and Design

New Thinking, Planning, & Operating

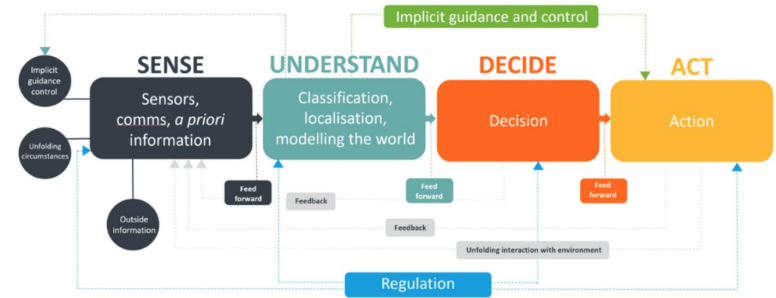
- Critical thinking has been the hallmark of industrial design and most organizations for over a century. Many companies have adopted design thinking to help reframe and explore ways they can introduce better products and services. Lots of innovation frameworks have also been introduced to enhance creativity, while most of these are derivative of DeBono's original work.
- Most organizations rely on decomposition for addressing complex and/or large problems. Componentization is borrowed from industrial design and has worked quite well for building scale.
- As the pace of change and complexity continue to accelerate, companies can't rely on deterministic thinking and planning. They need to embrace evolutionary thinking to survive in an expanding set of evolving (digital & physical) ecosystems.

From OODA to SUDA

The OODA loop was introduced in the 1950's by military strategist, US Air Force Colonel John Boyd to help pilots make faster and better decisions.

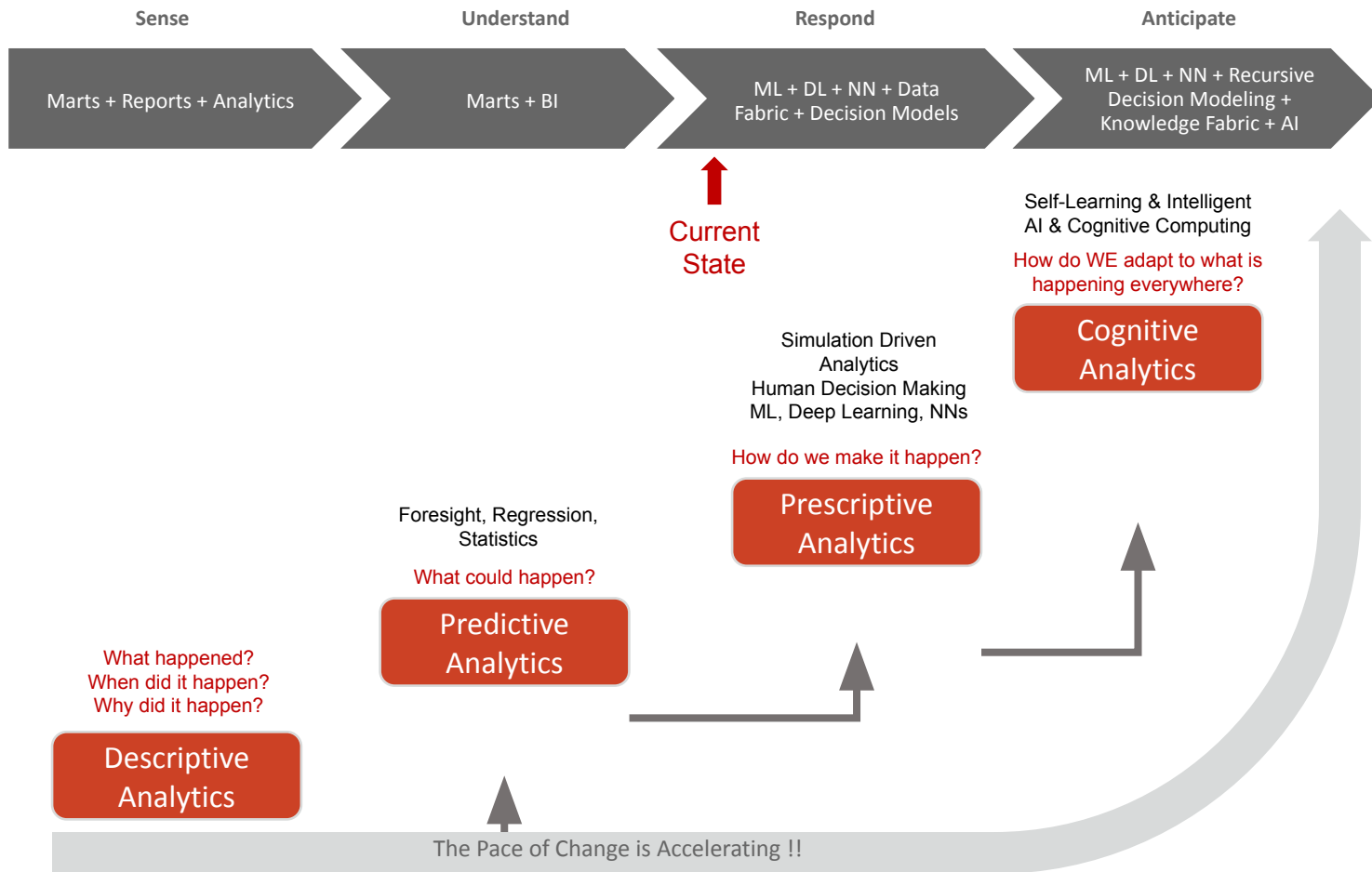


The SUDA model was developed more recently to improve decision making for AI by leveraging biomimicry to simulate the feedback & learning.



SUDA builds on OODA by introducing models of knowledge and mapping data (content & context) to concepts at scale under a governed framework of continuous learning and adaptation, which embraces complexity.

Roadmap to Achieve SUDA



Shift from app-centric to data-centric

App-Centric approach

- Start with the application
- Identify required sources
- Understand the data
- Align data to application
- Publish for use by the application
- Increment by business questions, requests

Data-Centric approach

- Start with business information model (design thinking)
- Identify a range of sources (broader semantic context)
- Understand the data
- Align the data to the information model
- Publish for use by many apps
- Increment by improving reusable data assets

Solutions are designed to leverage shared business-oriented data, contextually enriched and "naturally" linked together.

Data Centric Model

1. Consumption

- Users don't need to know how to negotiate different systems, technologies, or multiple languages. Introduces a semantic layer, which is the the pivot for enabling data-centricity and RT business consumption. Data contains the context and universal understanding to provide actionable knowledge to achieve business objectives.

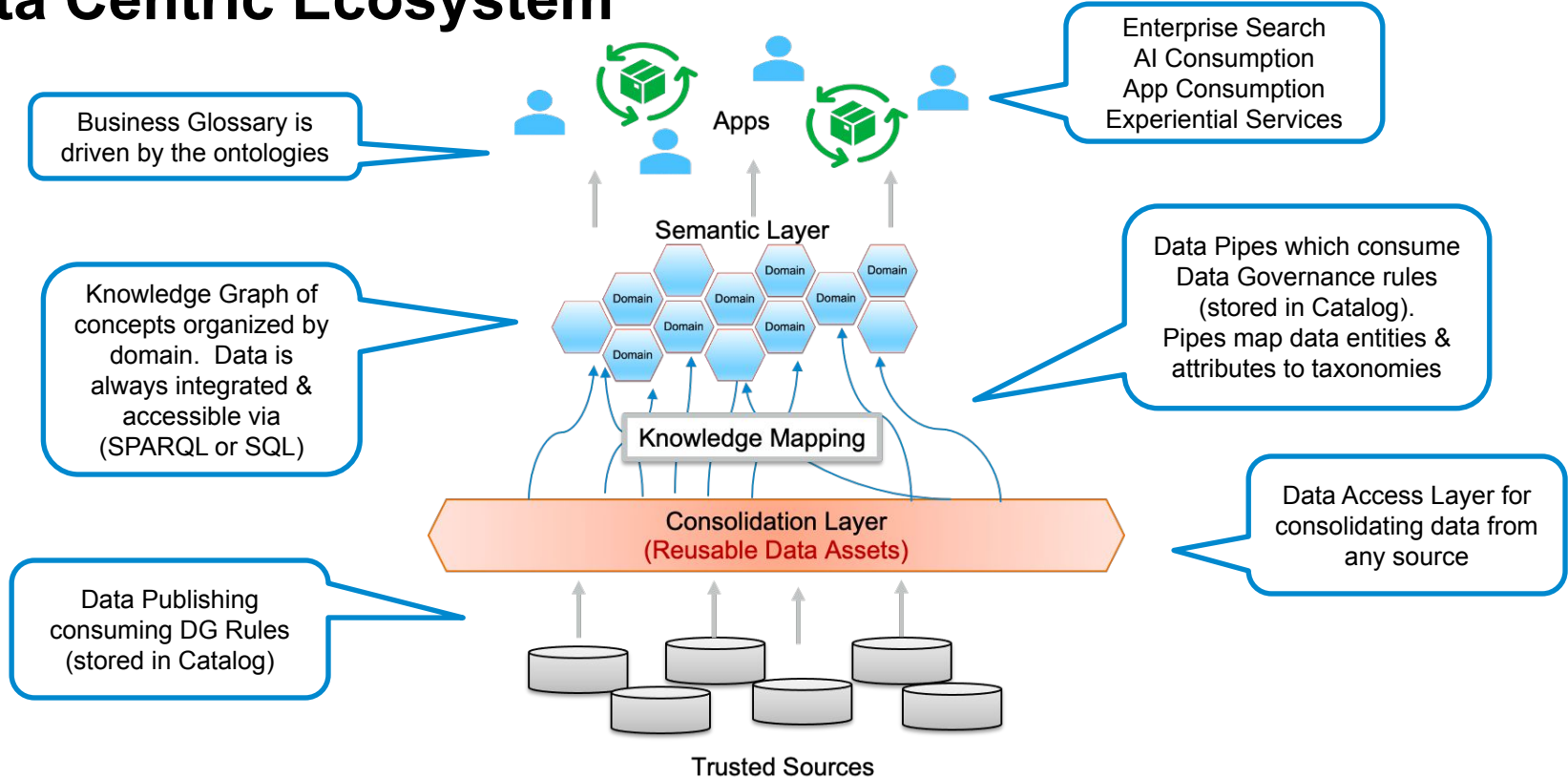
2. Operation

- This is where we can find trusted data presented as reusable assets (including data products). Provides democratized access to all data per Data Governance.

3. Generation

- All the data from the farms is harvested from the fields of silos and systems. Data is captured, cultivated, and consolidated. Data harvesting of content + context (= attributes) is enabled by an integration layer (TBD) which applies data governance (quality, security, policy).

Data Centric Ecosystem



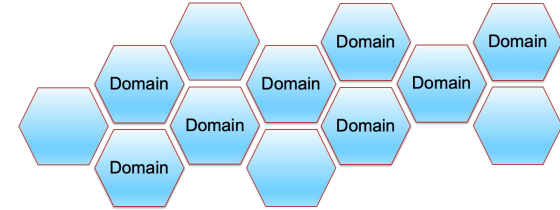
Semantic Layer (key differentiator)

Semantic Layer

- Graph & Relational Forms
- Interact with data using business terms
- Marketplace is organized in a way that does not require a manual or training. It's like walking into a grocery store.
- Like products are placed together (fruits, vegetables, breads, canned goods, etc.)
- Products are labeled consistently (don't need to translate)

Data Fabric

- Scaffolding of ontologies, services (functions, & processes), rules, and algorithms organized by domain
- Data and knowledge (content and context including semantics) are integrated between ontologies within domains and between domains
- Domains represent capabilities that flow vertically and horizontally across the organization. They are not crafted around the organizational structure or business processes.
- The domains support impact and insights spanning strategy, execution, and operations.



The fabric initially will have one layer, but additional layers will evolve to support layers of complexity and scale. Think of these as organic "Lego" blocks for supporting intuitive exploration, collaboration, and unified understanding.

Establishing Provider, Partner, & Consumer roles

- **Providers** deliver the data sources that are used to cultivate and curate data in the consolidation layer
- **Partners** collaborate on building and supporting the data supply chain to deliver consumable data and knowledge for consumers
- **Consumers** browse, sample, and subscribe to data services, compose business questions directly to the marketplace in real time.

Partners Include:

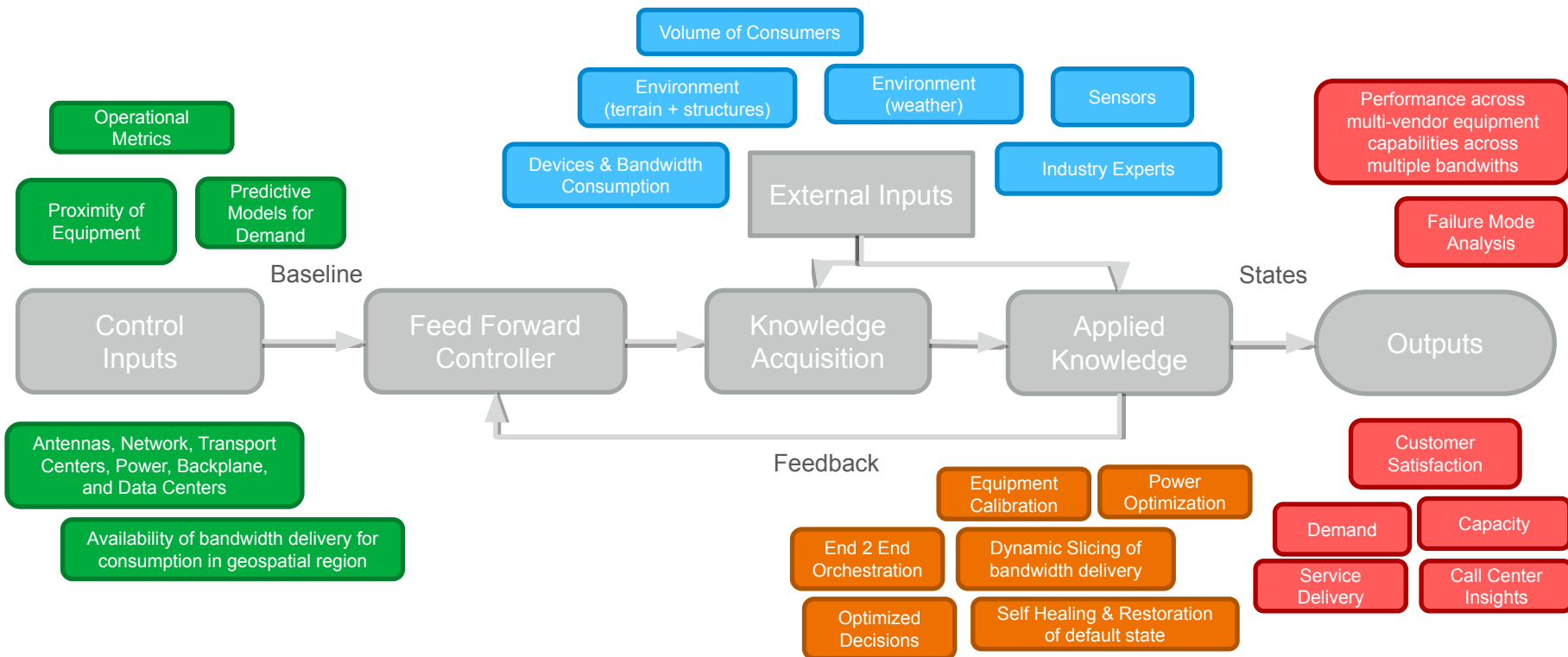
- Domain SMEs
- Knowledge Engineers (models: semantics, Tax, Ont)
- Data Engineer (data quality & pipelines)
- Governance (quality, security, policy)
- Data Steward (policies, quality, operations)
- Data Operations (customer support, SLAs, Op Ex)
- Data Scientists (AI, Model Consumption & Training)
- Data Investigations (RCA, Six Sigma, Cont. Improv.)
- Product Management (roadmap and strategy)

Applying Control Theory (SUDA) to Build Digital Twin Reference Model

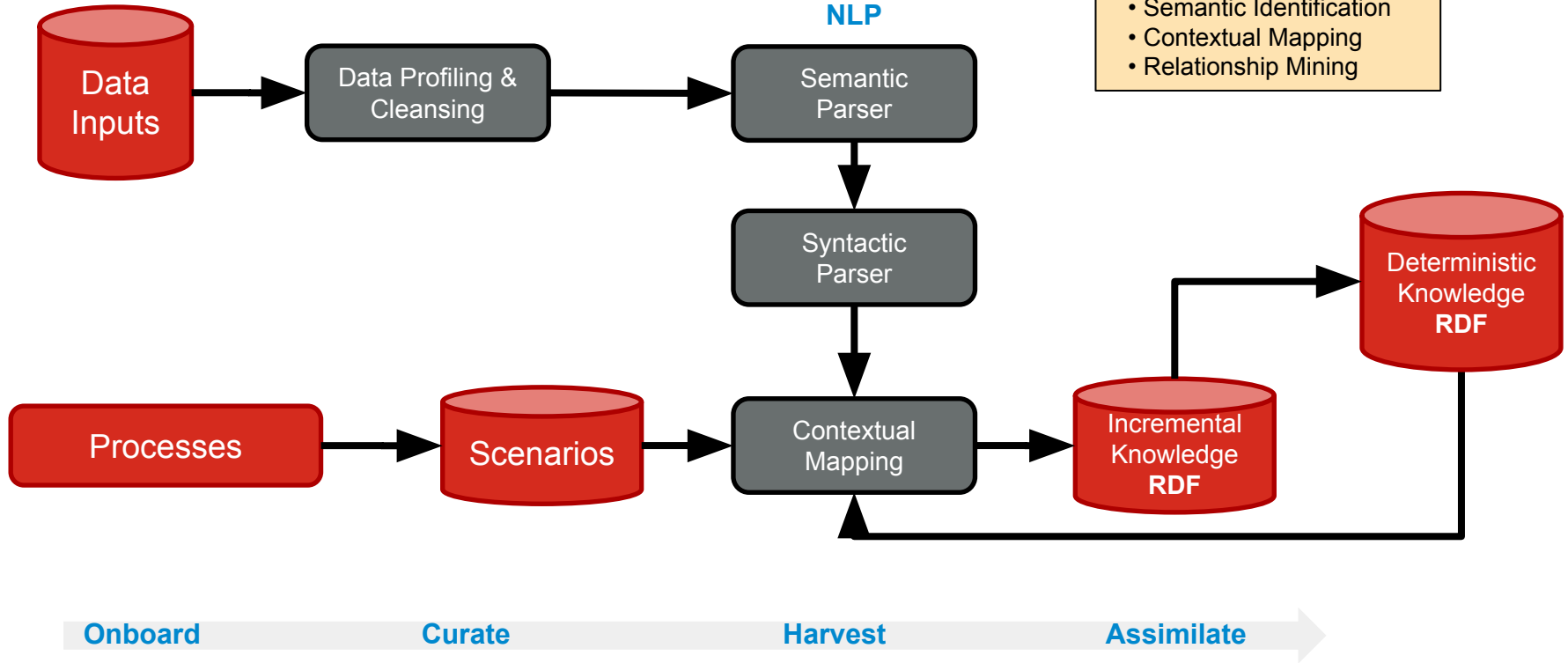
Key Benefits of KG adoption

- Combines Conceptual & Logical Views of data
- Preserves enterprise and community perspectives of “truth”
- Shifts spend on fixing existing problems so we can solve new ones
- Increases adaptability and resiliency for consumers of data & information
- Provides unlimited scale to manage volume and complexity
- Integrates the lifecycle and operational management (systems, products, services, data, analytics)
- Provides a foundation for designing and managing Digital Twins while improving AI and leveraging ML for knowledge capture & consumption
- Accelerates innovation and cycle time delivery for new products & services

(Control Theory + Theory of Constraints + Complexity Theory + Game Theory)



Knowledge Acquisition Pipeline



Assimilation

Assimilating incremental knowledge into Deterministic Knowledge (ontologies)

- an entirely new concept
- an existing concept with a new relation + scenario(s)
- an existing concept with a new attribute + scenario(s)

Insertion into Deterministic Knowledge (ontologies) requires evaluating

- Equality
- Equivalence
- Similarity

